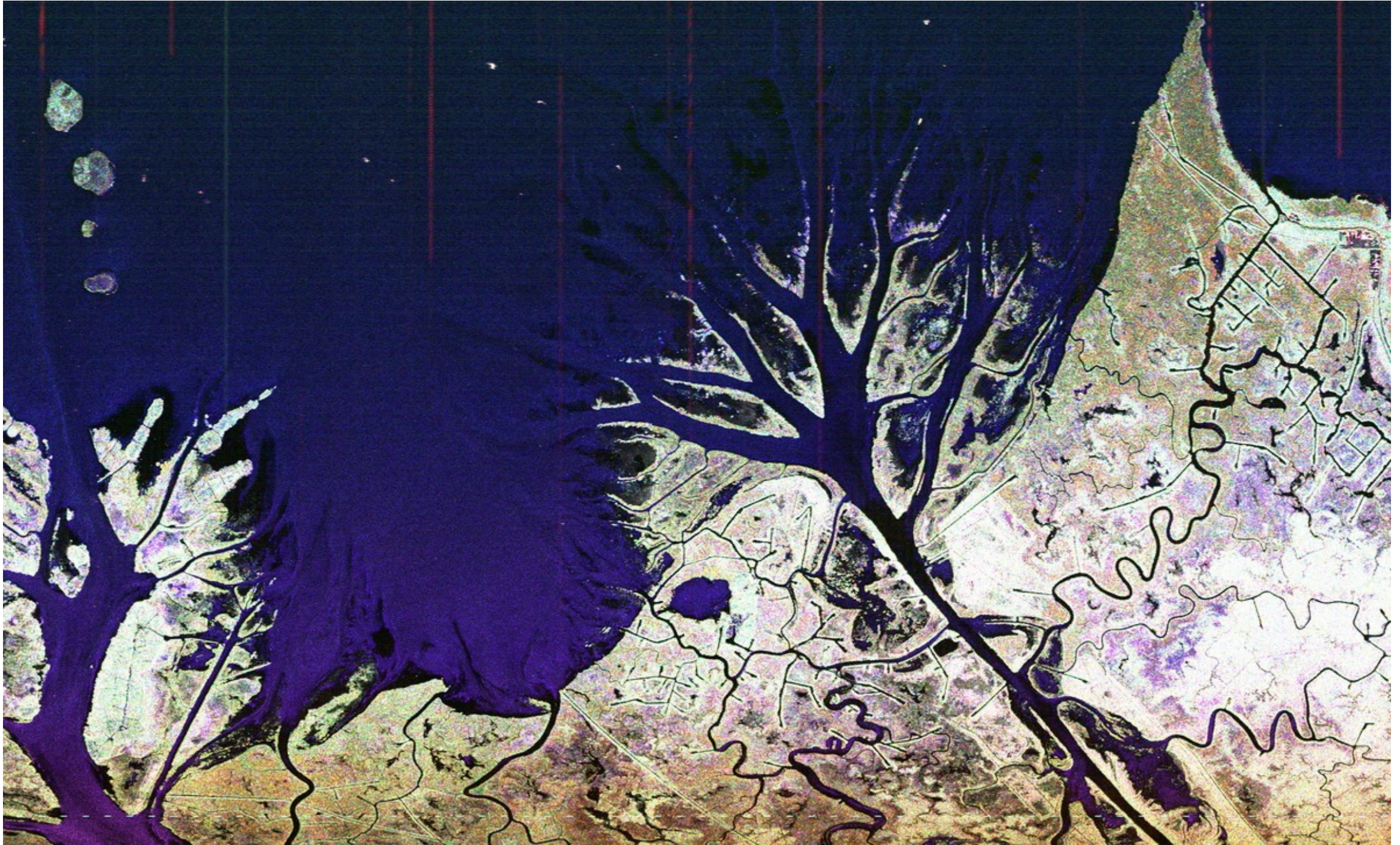


Lecture 1: Dielectric properties of materials

Physical principles of remote sensing
(Ge 158)
Fall 2026
Brent Minchew

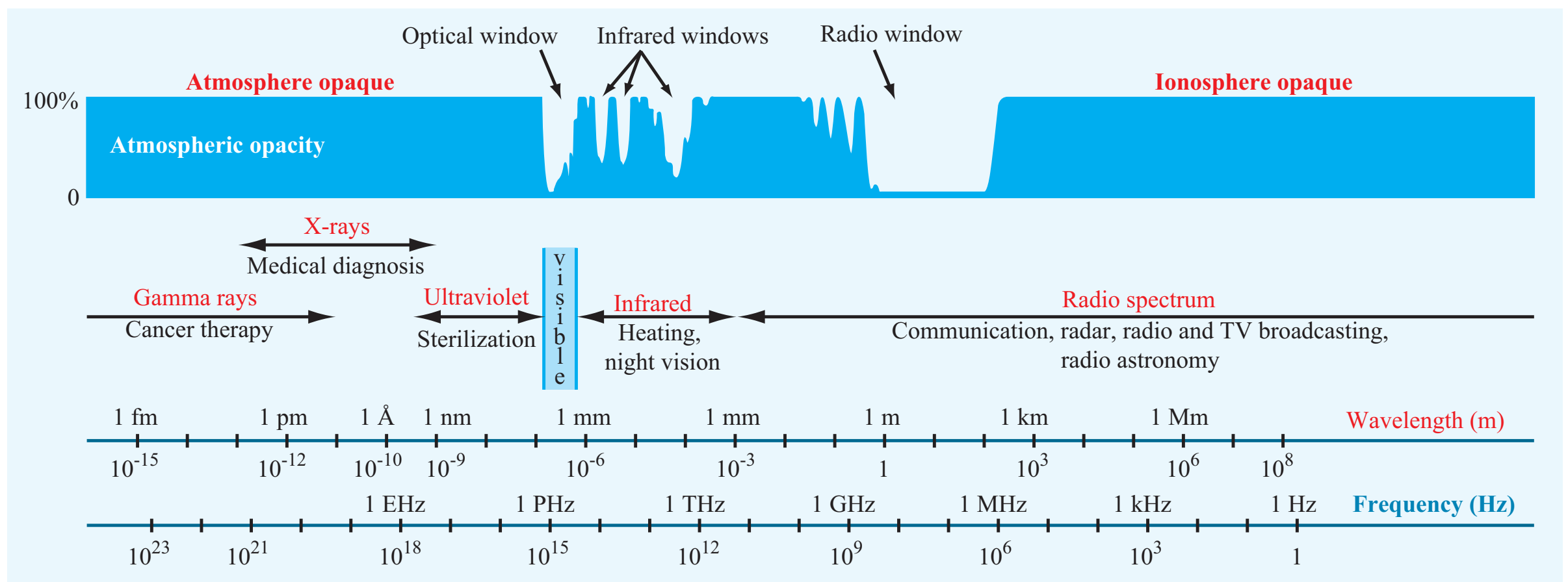
Remote sensing image of the day



Radar (UAVSAR) backscatter-power image of the Mississippi River Delta. Credit: C. Jones, NASA/JPL

Topics to be covered

- Constitutive parameters
- Permittivity and material polarization
- Relaxation mechanisms and simple dielectric models for pure water, salt water, and ice



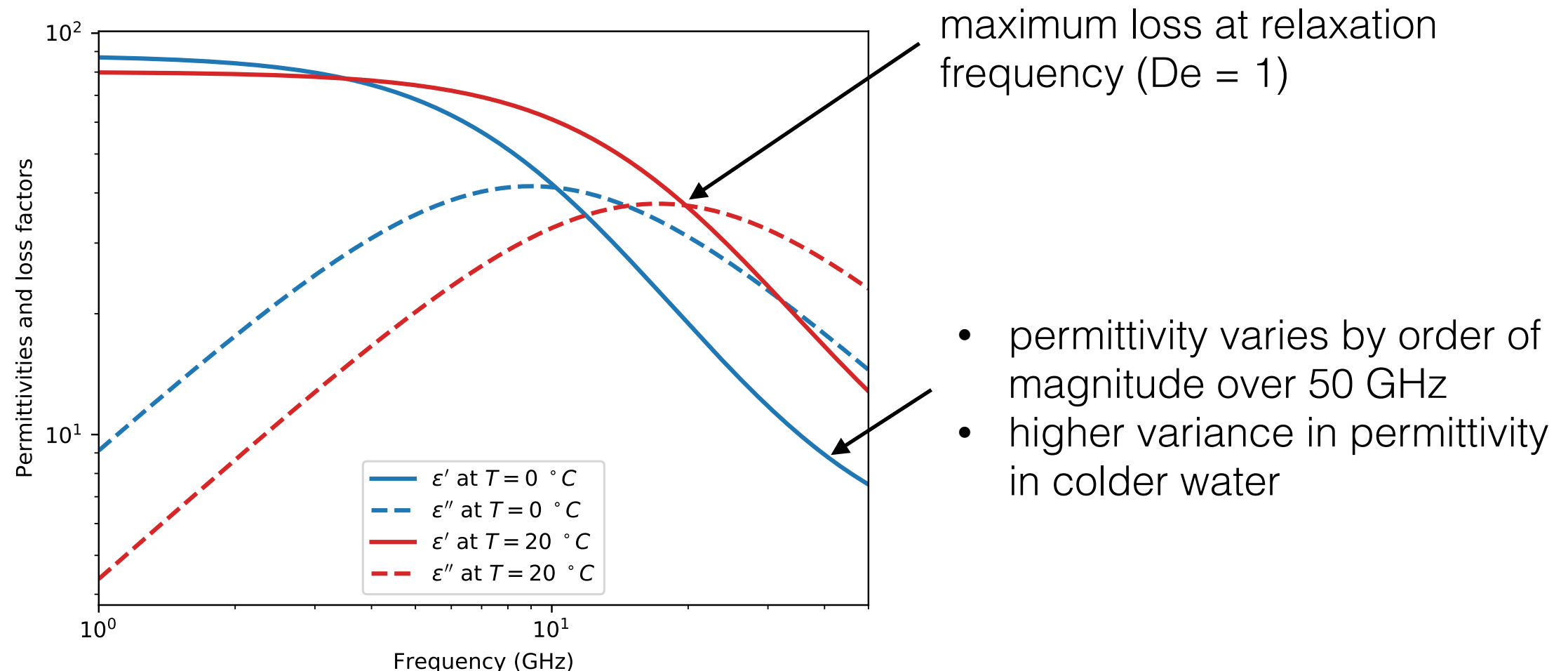
EM spectrum and atmospheric transparency, from Ulaby and Long, 2014

Dielectric constant of pure water, limited frequency range

- Debye relaxation (for a single relaxation mechanism)

$$\varepsilon' = \varepsilon_{w\infty} + \frac{\varepsilon_{w0} - \varepsilon_{w\infty}}{1 + \text{De}^2} \quad \text{and} \quad \varepsilon'' = \frac{\text{De} (\varepsilon_{w0} - \varepsilon_{w\infty})}{1 + \text{De}^2}$$
$$\text{De} = \omega\tau_w \quad \tau_w = \tau_w(T) \quad \varepsilon_{w0} = \varepsilon_{w0}(T) \quad \varepsilon_{w\infty} \approx 4.9$$

- At microwave frequencies ($f \leq 50$ GHz)
 - dipole orientation is the dominant relaxation mechanism
 - relaxation frequency within microwave spectrum



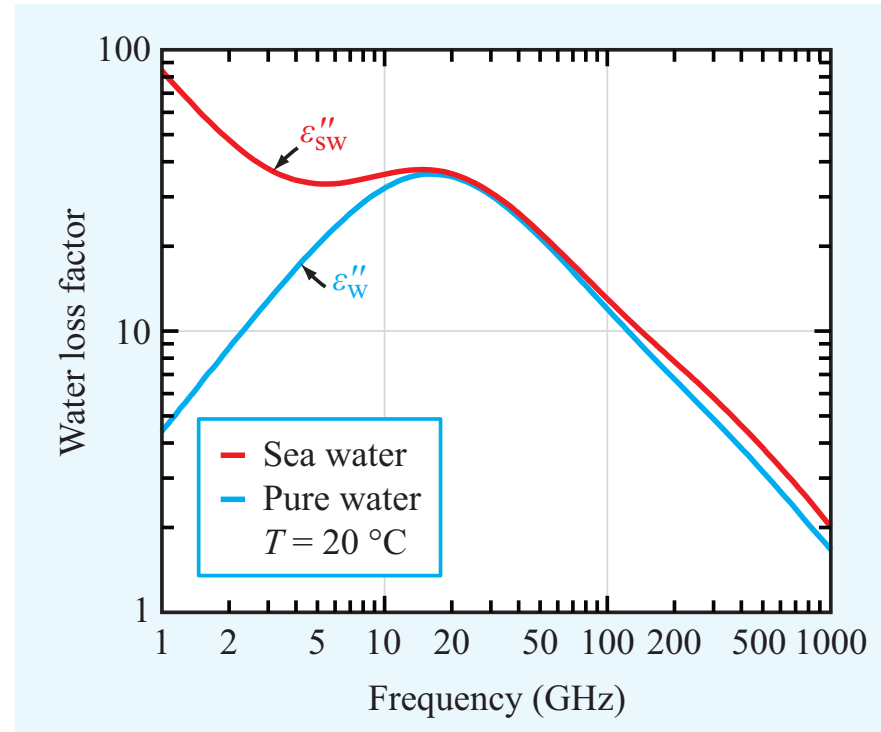
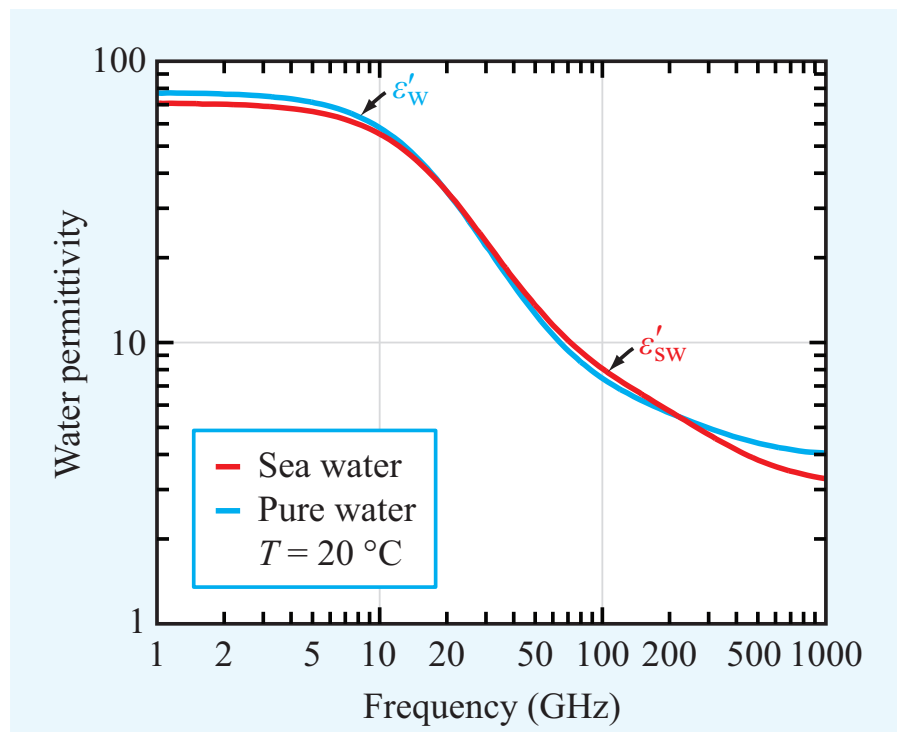
Dielectric constant of salt water, broader frequency range

- Debye relaxation model with two relaxation mechanisms (details omitted because they offer little intuition, involve ~20 fitted parameters, and are not essential for this course)

$$\epsilon' = \epsilon_{w\infty} + \frac{\epsilon_{w0} - \epsilon_{w1}}{1 + \text{De}_1^2} + \frac{\epsilon_{w0} - \epsilon_{w\infty}}{1 + \text{De}_2^2}$$
$$\epsilon'' = \frac{\text{De}_1 (\epsilon_{w0} - \epsilon_{w1})}{1 + \text{De}_1^2} + \frac{\text{De}_2 (\epsilon_{w0} - \epsilon_{w\infty})}{1 + \text{De}_2^2} + \frac{\sigma_i}{\omega \epsilon_0}$$

$$\text{De}_k = \omega \tau_{wk}(T, S) \text{ for } k = 1, 2, \quad \epsilon_{wk} = \epsilon_{wk}(T, S) \text{ for } k = 0, 1, \infty \quad \text{and} \quad \sigma_i = S\Upsilon(T, S)$$

- Additional relaxation mechanism extends frequency range into the infrared ($f \leq 1000$ GHz)
- Two relaxation mechanisms and ionic conduction (σ_i) make the model valid for salt water



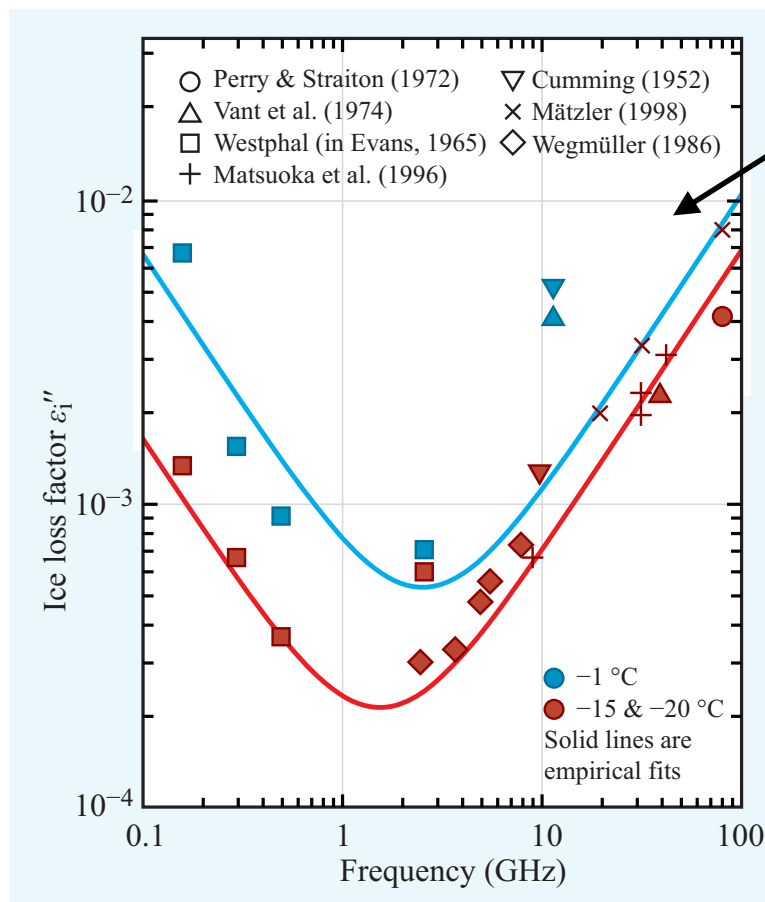
Dielectric constant of salt water at microwave to infrared frequencies for $T=20$ °C and $S = 32.5$ psu. Figure from Ulaby and Long (2014)

Dielectric constant of water ice

- Crystalline structure of water ice eliminates most relaxation mechanisms
- Reduces dielectric model to single relaxation frequency that is well within the radio region of the EM spectrum ($\sim$ kHz)
- For pure water ice over most practical frequencies in remote sensing, we have (details omitted)

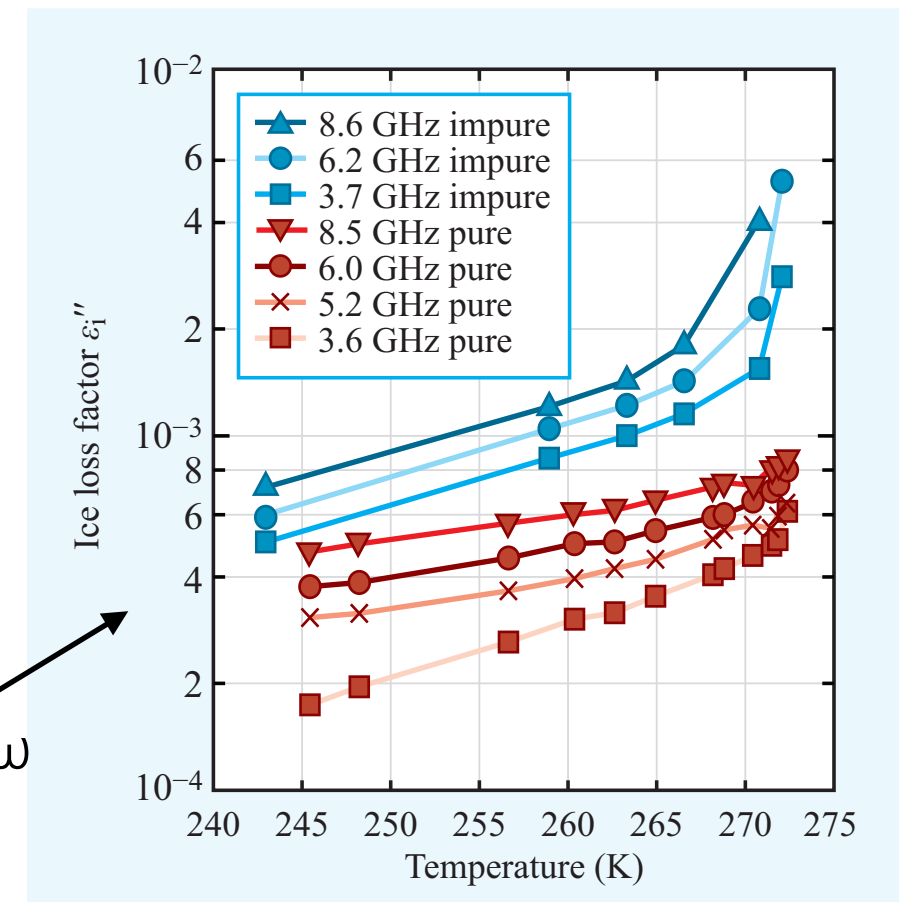
$$\epsilon' \approx 3.2 \quad \text{and} \quad \epsilon'' = \epsilon''(\omega, T)$$

- Note that the loss factor is 1-2 orders of magnitude smaller for pure ice than for pure water



effects of
T and ω

Loss factor for pure water ice at different temperatures and frequencies. From Ulaby and Long (2014)



effects of T, S, and ω

Loss factor for pure and impure water ice at different temperatures and frequencies. From Ulaby and Long (2014)

Other natural materials

- Broadest category of dielectric models encountered in RS will be heterogenous mixtures
- We will discuss selected heterogenous models if necessary later in the course
- But the an important point is that many mixtures involve geological materials (rocks, minerals, soils) moistened by liquid water
- Dry natural materials tend to have dielectric properties described as

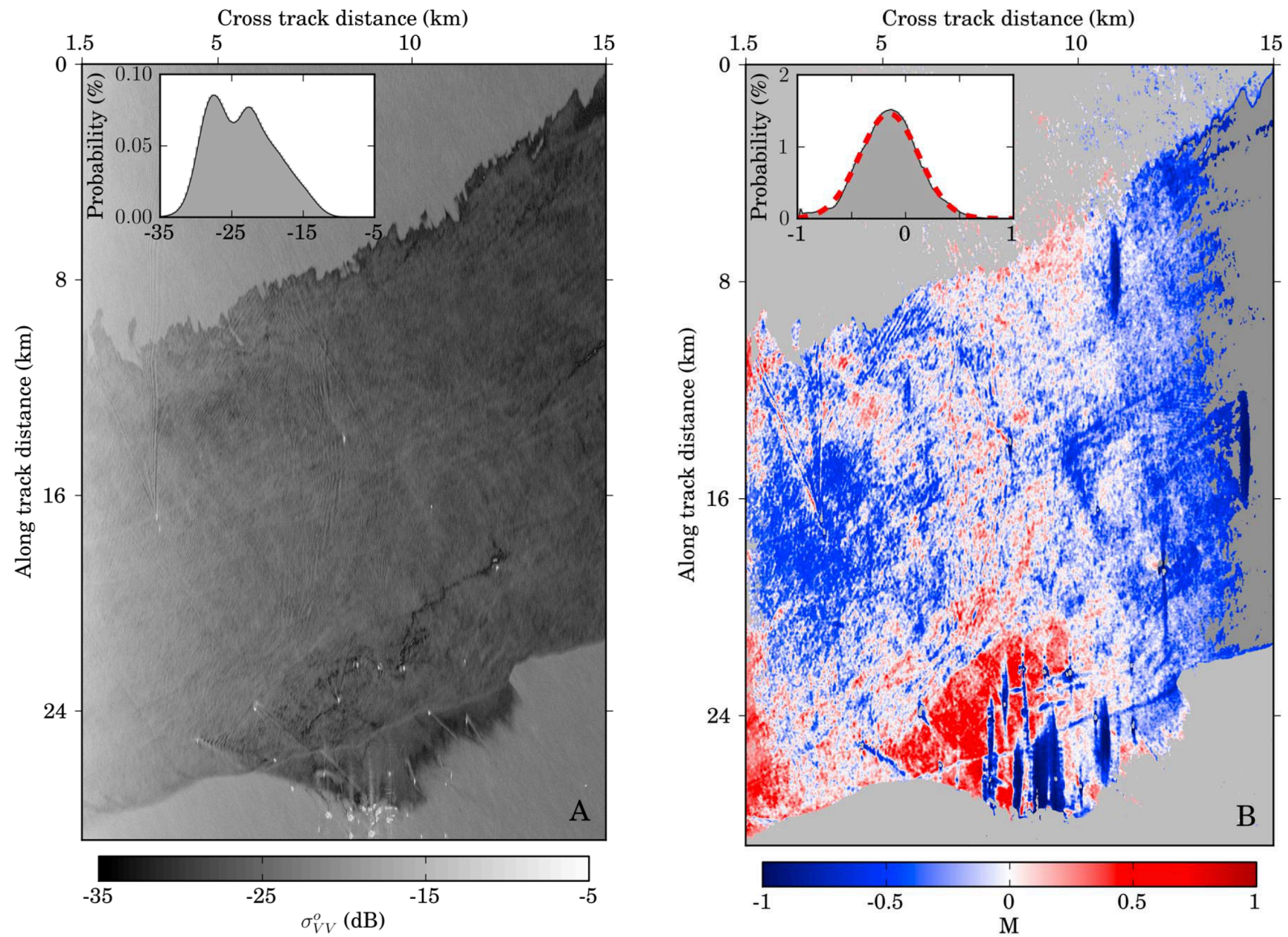
$$1 < \varepsilon' < 10 \quad \text{and} \quad \varepsilon'' < 0.1$$

compare with water

$$10 < \varepsilon'_{water} < 100 \quad \text{and} \quad 5 < \varepsilon''_{water} < 50$$

Therefore, moisture has a major influence on dielectric constant

RS applications of today's lecture



Minchew, 2012

Brief Summary (most material was covered in lecture)

- Two important constitutive parameters for RS: permittivity and conductivity, the latter governs attenuation
- A variety of mechanisms govern the frequency dependence of permittivity and conductivity for given material properties
- Properties like temperature, salinity, and phase can significantly influence permittivity and loss factor
- Presented simple models for pure water over limited frequency range (single relaxation model), a more complex model for salt water and broader freq. range (double relaxation model), and showed how crystalline structure of water ice alters the principal relaxation mechanism (reduces relaxation freq.), thereby changing the permittivity, loss factor, and their dependences on T and S